

Technical Assignment – Full Stack Developer

AI-Powered Document Reader

Objective

Develop an AI-powered Document Reader application that can extract, analyze, and answer questions from uploaded documents.

Functional Requirements

1. Upload PDF documents.
2. Extract text from uploaded PDFs.
3. Display extracted text.
4. Allow users to ask questions based on the document.
5. Generate answers using an LLM (Gemini/OpenAI or equivalent).
6. Maintain chat history for the current document.
7. Gracefully handle invalid or scanned PDFs (OCR is a bonus).
8. Provide meaningful error handling and logging.

Technical Requirements

- Frontend: React (Preferred)
- Backend: Python (FastAPI/Django/Flask)
- AI Integration: Gemini API or equivalent LLM
- Database: PostgreSQL/SQLite (if required)
- Version Control: Git
- Deployment (Bonus): Azure, AWS, or any cloud platform

Evaluation Criteria

- Project architecture
- Clean, modular, maintainable code
- Python coding standards
- API design
- AI/LLM integration
- Error handling
- Performance & scalability
- UI/UX
- Documentation

Deliverables

- GitHub repository link
- README with setup & execution steps
- Sample input documents
- Screenshots or short demo video
- API documentation (Swagger/Postman if applicable)
- Deployment URL (if deployed)

Submission Instructions

Share all deliverables in a single email or document. Ensure the repository is accessible and all setup instructions are complete.